

Course Code: EET202
Course Name: DC Machines & Transformers
(2019 Scheme)

ANSWER KEY

PART – A

1.

Let,

ϕ : Flux per pole in Weber (Wb)

Z : Total number of armature conductors

P : Number of poles

A : Number of parallel paths

N : Speed of armature in rpm

E_g : EMF generated in volts

Flux cut by one conductor in one revolution of armature, $d\phi = P\phi$

Time taken to complete one revolution, $dt = 60/N$ sec

EMF generated in one conductor, $\frac{d\phi}{dt} = \frac{P\phi}{60/N} = \frac{NP\phi}{60}$

So, total EMF generated, $E_g = (\text{EMF per conductor}) \times \text{number of conductors in one parallel path}$

$$= E_g = \frac{NP\phi}{60} \times \frac{Z}{A} = \frac{Z\phi N}{60} \times \frac{P}{A}$$

2. The dummy coils are similar to other coils except their ends are cut, short and taped. They do not connect with the commutator bars. The dummy coils are simply to provide mechanical balance for the armature.

$$P = 4$$

$$S = 24$$

$$\text{Conductors / slot} = 2$$

$$\text{Number of conductors, } Z = 24 \times 2 = \underline{\underline{48}}$$

$$\begin{aligned} \text{Winding pitch, } Y_w &= \frac{Z \pm 2}{P/2} \\ &= \frac{48 \pm 2}{4/2} \end{aligned}$$

$$= \frac{50}{2} = \underline{25} \text{ (assume progressive winding)}$$

$$Y_w = Y_b + Y_F$$

For wave winding Y_w should be an even number, then only Y_b and Y_F can be odd.

Since here winding pitch, (Y_w) is odd, dummy coils should be introduced.

Thus out of ~~50~~ ⁴⁸ conductors, 2 conductors should be used as dummy coils.

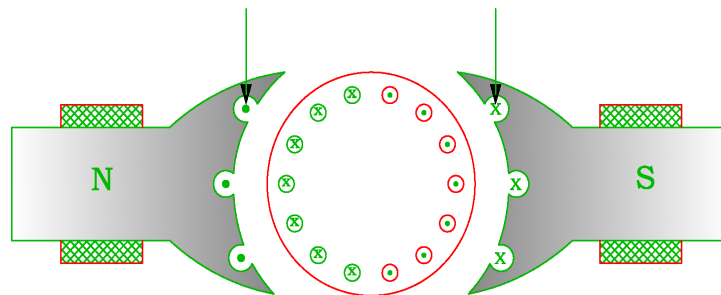
Then, $Z = 46$

$$Y_w = \frac{46 + 2}{4/2} = \underline{24} \text{ (even number)}$$

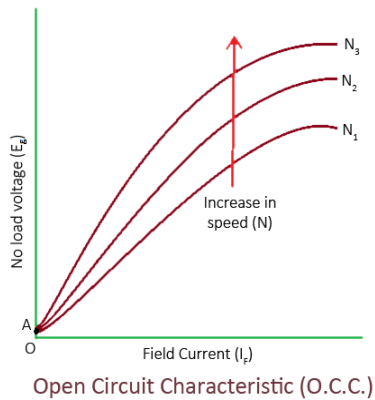
3.

- The cross-magnetising effect leads to sparking at the brushes.
- In order to neutralise cross-magnetising effect of armature reaction compensating windings are used.
- Compensating windings are the auxiliary windings embedded in slots in the pole faces.
- They are connected in series with armature in a manner so that the direction of current through compensating conductors in any one pole face will be opposite to the current through the adjacent armature conductors.

COMPANSATING WINDING



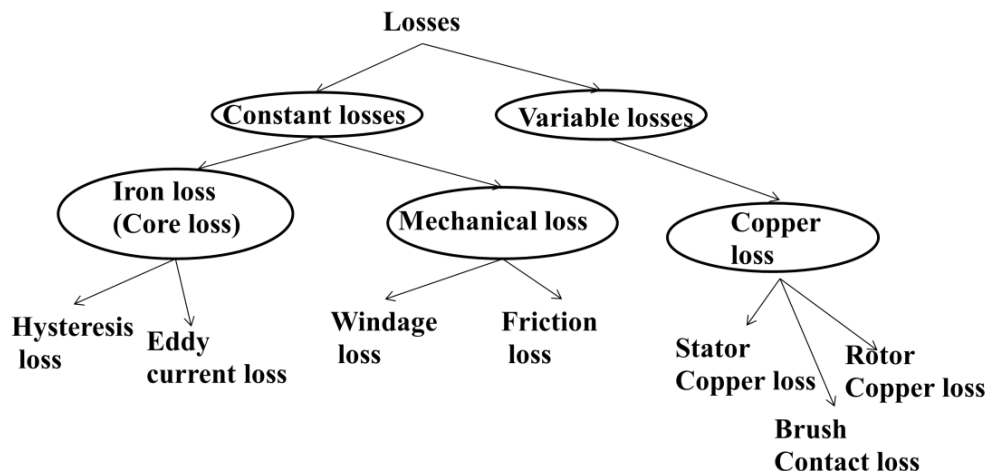
4.



Causes for the failure of the machine to build up voltage

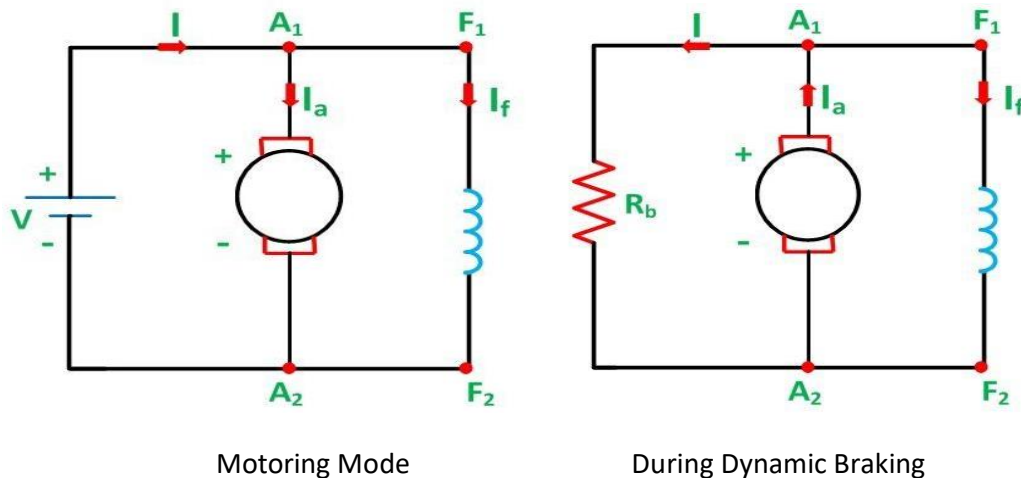
- **No residual magnetism:** The voltage build up procedure can start only if there is residual magnetism. In case of new machine or the one that has lost its magnetism due to aging, self excited generator will fail to build up the voltage. This difficulty can be overcome by exciting the field winding from a separate DC source for a few seconds with the armature at rest. This process is called *Field Flashing*.
- **Field connections reversed:** The small voltage due to residual magnetism, should circulate current in the field coils in such a direction as to produce flux lines aiding the residual flux. If the field connections are incorrect, the flux produced by the small field current opposes the residual flux and the generated voltage decreases to zero. This trouble can be remedied by reversing the field connections with respect to armature terminals.
- **High field circuit resistance:** In case the field resistance is more than the critical field resistance, voltage will not build up.
- **Speed less than critical speed:** Generator may fail to build up the voltage, due to the armature speed being less than the critical speed. This trouble can be remedied by increasing the prime-mover speed above critical speed.

5.



6. Electric braking methods
 - a. Regenerative Braking
 - b. Dynamic Braking
 - c. Plugging

In Dynamic Braking, a braking resistor R_b is connected across the armature as soon as the DC motor is disconnected from the supply mains. The motor now works as a generator, producing the braking torque.



Connection diagram for dynamic braking is shown above.

During electric braking, the kinetic energy stored in the rotating parts of the machine and the connected load is converted into electric energy, when the motor is working as a generator. The energy is dissipated as heat in the braking resistance R_b and armature circuit resistance R_a .

The Dynamic or Rheostatic Braking is an insufficient method of braking because all the energy which is generated is dissipated in the form of heat in the resistance.

7.

$$\text{Efficiency, } \eta = \frac{\text{Output}}{\text{Input}} = \frac{\text{Input} - \text{Losses}}{\text{Input}}$$

Losses can be found out from OC and SC tests.

OC Test: The wattmeter reading (W_c) in OC tests gives the core losses (Iron losses) at rated voltage.

SC Test: The wattmeter reading (W_{cu}) in SC test gives the copper losses at full load condition.

To determine Losses at any load condition:

Core losses are independent of load.

Thus, Core Losses at any load condition = W_c

Copper losses at any loaded condition

If the transformer loaded condition is 'x' times the full load,

Copper losses = $x^2 W_{cu}$

Thus,

$$\eta = \frac{\text{Input} - \text{Losses}}{\text{Input}} = \frac{\text{Input} - (W_c + x^2 W_{cu})}{\text{Input}}$$

8.

$$\text{Cu losses} = I_1^2 r_{01} = I_2^2 r_{02}$$

$$\text{Iron losses} = W_i$$

$$\begin{aligned} \text{Efficiency, } \eta &= \frac{\text{Output}}{\text{Input}} = \frac{\text{Input} - \text{Losses}}{\text{Input}} \\ &= \frac{V_1 I_1 \cos \phi_1 - I_1^2 r_{01} - W_i}{V_1 I_1 \cos \phi_1} \\ &= \frac{V_1 I_1 \cos \phi_1}{V_1 I_1 \cos \phi_1} - \frac{I_1^2 r_{01}}{V_1 I_1 \cos \phi_1} - \frac{W_i}{V_1 I_1 \cos \phi_1} \\ &= 1 - \frac{I_1 r_{01}}{V_1 \cos \phi_1} - \frac{W_i}{V_1 I_1 \cos \phi_1} \end{aligned}$$

Differentiating with respect to I_1

$$\frac{d\eta}{dI_1} = 0 - \frac{r_{01}}{V_1 \cos \phi_1} + \frac{W_i}{V_1 I_1^2 \cos \phi_1}$$

For Maximum Efficiency,

$$\begin{aligned} \frac{d\eta}{dI_1} &= 0 \\ \frac{r_{01}}{V_1 \cos \phi_1} &= \frac{W_i}{V_1 I_1^2 \cos \phi_1} \end{aligned}$$

$$I_1^2 r_{01} = W_i$$

$$\text{Cu loss} = \text{Core loss}$$

9.

Power Transformer	Distribution Transformer
It is used in transmission network of higher voltages 33KV and higher	It is used in the distribution network of lower voltages 11KV and lower
Used for rating above 200 MVA	Used for rating below 200 MVA
Larger in size as compared of distribution transformers	Smaller in size
Used in generating stations and transmission substations	Used in distribution stations, also for industrial and domestic purposes
Always operated at full load	Operated at load less than full load as load cycle fluctuates
Efficiency is measured as the ratio of output to the input power	All day efficiency is considered
Designed for maximum efficiency of 100%	Designed for 50-70% efficiency

10.

Various vector groups are:

- Yy0
- Dd0
- Dy1
- Yd1
- Dy11
- Yd11
- Dd6
- Yy6

PART – B

11.

a)

$$P = 4$$

$$A = 4$$

$$Z = 32$$

For lap winding,

$$Y_b = \frac{Z}{P} \pm K$$

$$= \frac{32}{4} \pm K$$

$$= 8 + 1$$

$$= \underline{\underline{9}}$$

$$Y_w = 2$$

$$Y_F = Y_b - Y_w$$

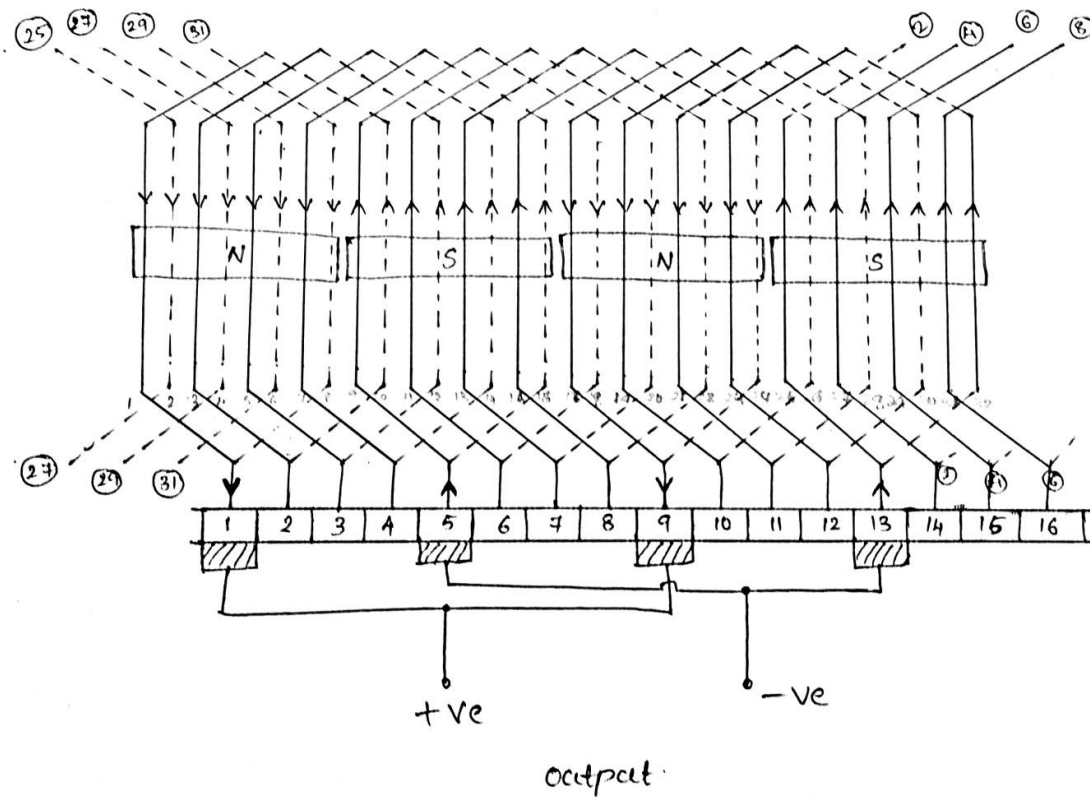
$$= \underline{\underline{7}}$$

Connection chart

Back Connection	Front Connection
$1 + 9 = 10$	$10 - 7 = 3$
$3 + 9 = 12$	$12 - 7 = 5$
$5 + 9 = 14$	$14 - 7 = 7$
$7 + 9 = 16$	$16 - 7 = 9$
$9 + 9 = 18$	$18 - 7 = 11$
$11 + 9 = 20$	$20 - 7 = 13$
$13 + 9 = 22$	$22 - 7 = 15$
$15 + 9 = 24$	$24 - 7 = 17$
$17 + 9 = 26$	$26 - 7 = 19$
$19 + 9 = 28$	$28 - 7 = 21$
$21 + 9 = 30$	$30 - 7 = 23$
$23 + 9 = 32$	$32 - 7 = 25$
$25 + 9 = 34(2)$	$34 - 7 = 27$
$27 + 9 = 36(4)$	$36 - 7 = 29$
$29 + 9 = 38(6)$	$38 - 7 = 31$
$31 + 9 = 40(8)$	$40 - 7 = 33(1)$

$$\text{Number of conductors / pole} = Z/p = \underline{\underline{8}}$$

Winding diagram is shown below:



- b) In lap winding, the number of parallel paths is equal to the total number of poles and is used in low voltage and high current dc machines.

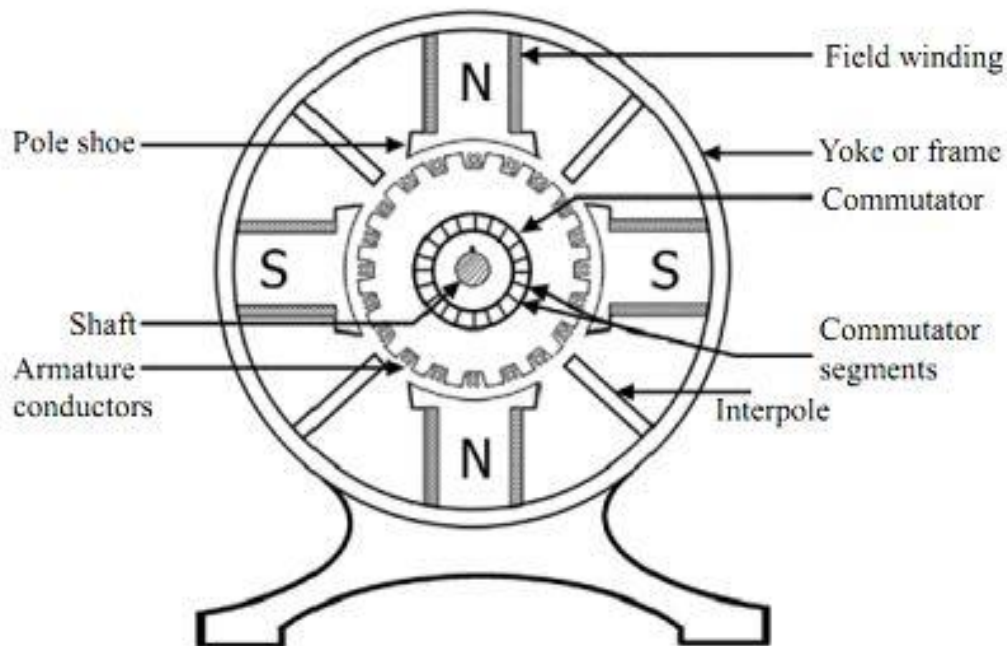
Under ideal conditions, the flux from all the poles is exactly same, hence the induced emf and current carried by each path will be equal. But under practical operating condition there is variations in air-gap length or difference in magnetic properties of poles material which causes a slight unbalance of induced emf in the various parallel paths.

As a result, conductor under stronger poles generates greater emf and carries larger current than remaining conductors. It means that some brushes are overloaded and carry more current than normal current while the remaining brushes carry less current due to under loaded. The current doesn't reverse completely in a given commutation time period due to overloaded brush and that will cause sparking between brushes and commutator surface. The equalizer rings are used to overcome such problem.

But in case of wave winding, there is only 2 parallel paths for any number of poles. Thus both the parallel paths are distributed under all the poles. Thus there will not be the issue of unequal current distribution and no need of equalizer rings.

12.

a)



Essential Parts are:

- Yoke or Frame
- Field System
 - Pole Core
 - Pole Shoe
 - Field Coils / Field Windings
- Armature
 - Armature Core
 - Armature Windings
- Commutator
- Brushes

Yoke or Frame: It is the outer cover of machine.

Functions:

- It serves the purpose of outermost cover of the dc machine. So that machine parts get protected from moisture, dust etc.
- Provide mechanical support to the poles.
- It provides a path of low reluctance for the magnetic flux produced by the poles.

Construction material:

It must be made up of some magnetic material to provide a low reluctance path for magnetic flux.

- In small machines cast iron is used for the construction because it is cheap. For large machines rolled steel, cast steel etc are used.

Pole Core: Every machine has even number of poles. Pole cores are bolted to the yoke.

Functions:

Pole core basically carries field winding which is necessary to produce flux.

Construction material:

Pole core is made of steel laminations

Pole Shoe: Each pole core has a pole shoe of curved surface.

Functions:

- Pole shoes support the field windings
- Spread out the flux in air gap uniformly
- Increase the cross-sectional area of the magnetic circuit and thus reduce the reluctance of the flux path.

Construction material:

- Pole shoe is made of steel laminations

Field Winding: Field windings are wound on the pole core to produce magnetic field.

They are also called pole coils or exciting coils. These coils are wound over the poles in such a direction so as to form alternate north (N) and south (S) poles.

Functions:

- In dc machines electromagnets are used for producing necessary flux. Whenever these field windings carry current, pole cores get electro-magnetised and produce flux.

Construction material:

- Copper is used for the construction of field windings.

Armature Core: It is cylindrical in shape and mounted on the shaft. It consists of slots on its outer periphery.

Functions:

- It houses armature windings on the slots provided on its periphery and causes them to rotate in the magnetic field.
- It provides a path of low reluctance for the magnetic flux produced by the filed system.

Construction material:

As it has to provide low reluctance path for the flux, it is made of cast steel. Armature core is made of laminations to reduce eddy current loss.

Armature Windings: These are placed in the slots provided on the periphery of armature core. Armature windings are of two types: Lap winding and Wave winding. Their selection will depend on the current rating of the machine.

Functions:

- In case of generator, generation of EMF takes place in the armature windings.
- In case of motors, it carries the supply current.

Construction material:

Armature windings are made of Copper.

Commutator: The commutator is of cylindrical in shape and divided into number of segments. Number of segments is equal to the number of armature coils.

Functions:

- Main function of the commutator is to convert alternating current induced in the armature into unidirectional current.

Construction material:

Commutator is made of wedge-shaped hard drawn forged copper. Commutator segments are insulated from each other with thin layer of mica sheet.

Brushes: The brushes are placed in brush holders. They rest on the commutator segments. The commutator and brush arrangement is the connecting link between the armature winding and external circuit.

Functions:

- Collect the current from the commutator of the machine

Construction material:

Brushes are made of Carbon.

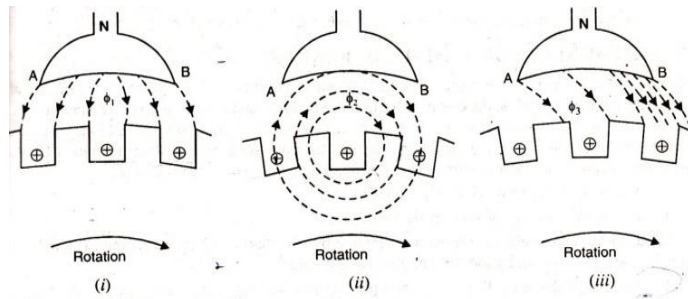
- b) In case of DC generator, the commutator is used to convert generated AC in armature to DC in external circuit. The physical connection to armature is done through a commutator-brush arrangement. The function of commutator in DC generator is to collect the current generated in armature conductors.

13.

The flux acting on dc machine which is developed by main poles is called main flux. When current flowing through armature conductors, it also creates magnetic flux and it is called armature flux.

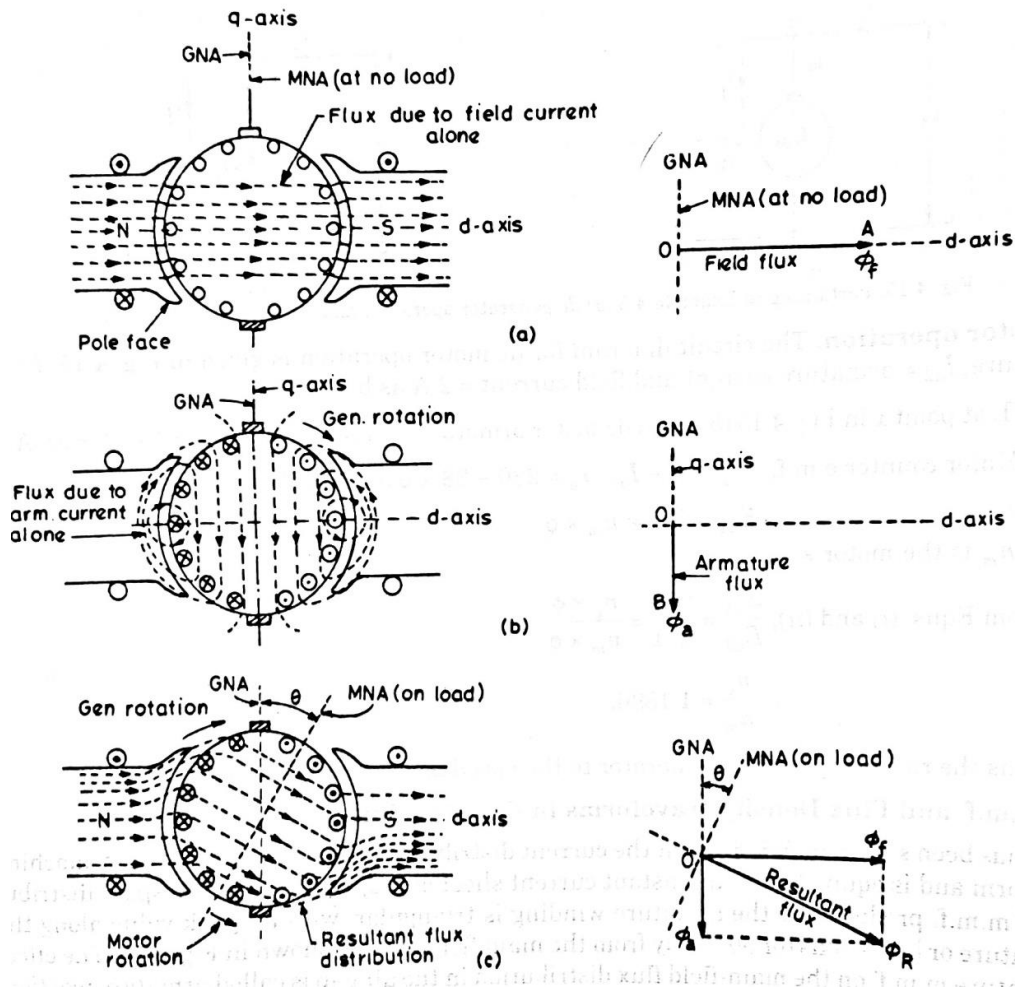
This armature flux **distorts and weakens** the main flux.

The action or effect of armature flux on main flux is known as armature reaction.



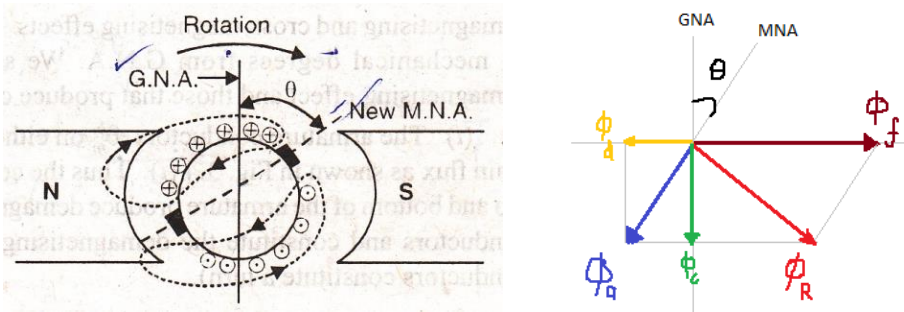
Major effects of armature reaction in DC machine

- Distorting the main flux (Cross-magnetisation)
- Weakening the main flux (demagnetisation)



- With no current in armature conductors, the flux across the air-gap is uniform and MNA coincides with GNA.
- When current flows in armature conductors, the combined action of main flux and armature flux (armature reaction) shifts MNA by an angle θ from GNA.
- In case of generator, MNA shifts in the direction of rotation of the machine.

- In order to achieve sparkless commutation, the brushes have to be moved along the new MNA.
- Due to the brush shift, armature mmf is also shifted by the same angle θ .
- The result is that armature mmf ϕ_a will no longer be vertically downward but will be rotated in the direction of rotation of generator by an angle θ .
- Now ϕ_a can be resolved in to 2 rectangular components ϕ_c and ϕ_d
- Component ϕ_d is in direct opposition to the main mmf. So it has a demagnetising effect on main flux and it is called **demagnetising (weakening) component of armature reaction**.
- The component ϕ_c is perpendicular to the main mmf. So it distort the main flux. Hence it is called **cross-magnetising (distorting) component of armature reaction**.
- The demagnetising effect leads to reduced generated voltage while cross-magnetising effect leads to sparking at the brushes.



Calculation of Demagnetising AT/pole

The effect of demagnetising Ampere-Turns(AT) can be neutralized by adding extra AT to the main field windings.

Let, Z : Total number of armature conductors

I : Current in each armature conductor

$$I = I_a / 2 \quad \text{for wave winding}$$

$$I = I_a / 4 \quad \text{for lap winding}$$

$$\text{Number of total demagnetising conductors} = \frac{4\theta_m}{360} \times Z$$

$$\text{Total demagnetising AT} = \frac{1}{2} \left[\frac{4\theta_m}{360} \times Z \right] I = \frac{2\theta_m}{360} \times ZI$$

These AT are due to a pair of poles.

$$\text{So demagnetising Ampere-Turns/pole, } AT_d/\text{pole} = \frac{\theta_m}{360} \times ZI$$

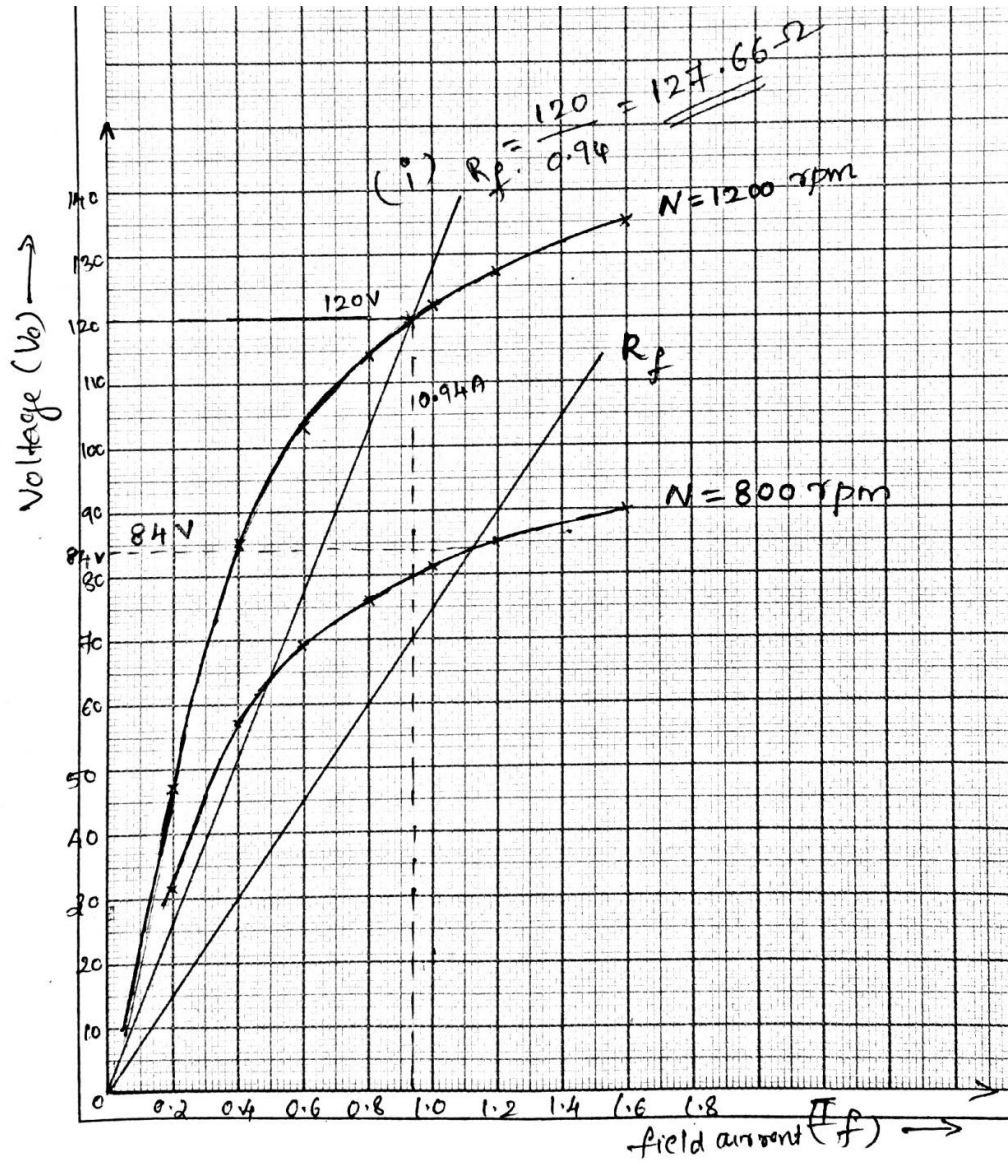
Calculation of Cross-magnetising AT/pole

$$\text{Armature AT/pole} = \frac{Z/2}{P} \times I = \frac{ZI}{2P}$$

$$\text{Total demagnetising AT} = \frac{\theta_m}{360} \times ZI$$

So cross-magnetising Ampere-Turns/pole, $AT_c/\text{pole} = \left[\frac{1}{2P} - \frac{\theta_m}{360} \right] ZI$

14.



(i) Field resistance, $R_f = \frac{120}{0.94} = 127.66\Omega$

Thus, field regulating resistance = $127.66 - 55 = 72.66\Omega$

(ii) To plot OCC at 800 rpm

$$\frac{E_1}{E_2} = \frac{N_1}{N_2}$$

$$E_2 = E_1 \times \frac{800}{1200}$$

I_f (A)	0.2	0.4	0.6	0.8	1.0	1.2	1.6
V_o (Volt)	31.33	56.67	68.67	76.0	81.33	84.67	90.0

Total field resistance with field regulator at $20\Omega = 55 + 20 = 75\Omega$

Open cir voltage = 84 V

15.

a)

Torque developed by the armature,

$$\tau_e = 0.159 \frac{Z\phi I_a P}{A}$$

$$Z = 120 \times 8 = 960$$

$$\phi = 0.02 \text{ Wb}$$

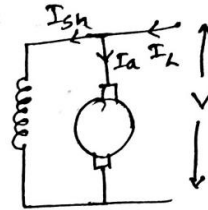
$$P = 4 = A$$

$$I_a = I_L - I_{sh}$$

$$I_L = 25 \text{ A}$$

$$I_{sh} = \frac{V}{R_{sh}} = \frac{250}{125} = 2 \text{ A}$$

$$\therefore I_a = 25 - 2 = \underline{\underline{23 \text{ A}}}$$



$$\tau_e = 0.159 \times 960 \times 0.02 \times 23 \times \frac{4}{4}$$

$$= \underline{\underline{70.214 \text{ Nm}}}$$

$\therefore$ Power developed in armature, $P_a = \tau_e \omega$

$$= \tau_e \frac{2\pi N}{60}$$

To find speed (N)

$$E_b = V - I_a R_a = 250 - (23 \times 0.1) = \underline{\underline{247.7 \text{ V}}}$$

$$E_b = \frac{Z\phi N}{60} \times \frac{P}{A}$$

$$N = \frac{E_b \times 60}{Z\phi} \times \frac{A}{P} = \underline{\underline{774.06 \text{ rpm}}}$$

$$\therefore P_a = T_e \frac{2\pi N}{60}$$

$$= \underline{\underline{5691.52 \text{ W}}}$$

$$\text{rotational losses} = 810 \text{ W}$$

$$\therefore P_{out} = P_a - P_{rot}$$

$$= 5691.52 - 810 = \underline{\underline{4881.52 \text{ W}}}$$

$$\therefore \text{Useful torque} = \frac{P_{out}}{\omega} = \frac{P_{out}}{(\frac{2\pi N}{60})}$$

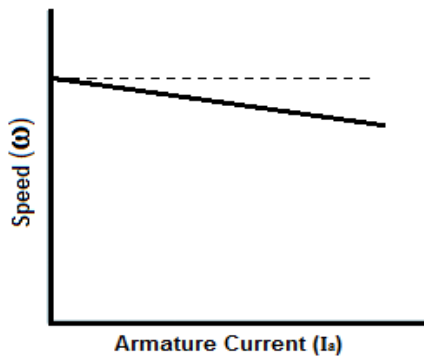
$$= \underline{\underline{60.22 \text{ Nm}}}$$

b)

- **Speed-Current characteristics (ω vs I_a):** It shows the relation between motor speed and armature current for constant applied voltage.
- **Torque-Current characteristics (T_a vs I_a):** It shows the relation between torque developed in the armature with armature current for the constant applied voltage.
- **Speed-Torque characteristics (ω vs T_a):** It shows the relation between speed and the torque developed by the motor for constant applied voltage.

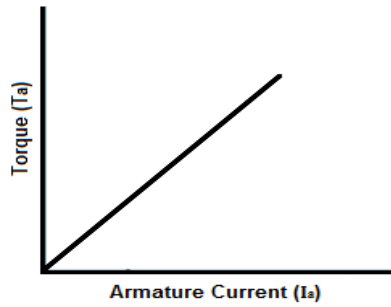
SHUNT MOTOR

Speed-Current characteristics (ω vs I_a)



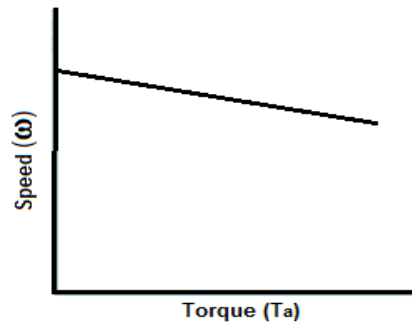
$$\omega = \frac{V}{K\phi} - \frac{R_a}{K\phi} I_a$$

Torque-Current characteristics (T_a vs I_a)



$$T_a = K\phi I_a$$

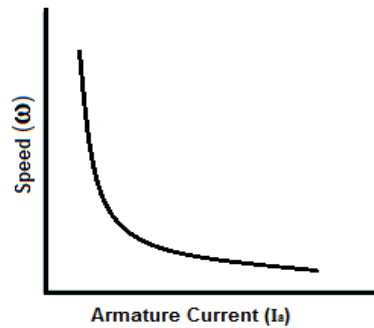
Speed-Torque characteristics (ω vs T_a)



$$\omega = \frac{V}{K\phi} - \frac{R_a}{K^2\phi^2} T_a$$

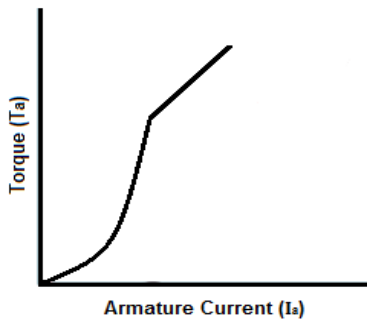
SERIES MOTOR

Speed-Current characteristics (ω vs I_a)



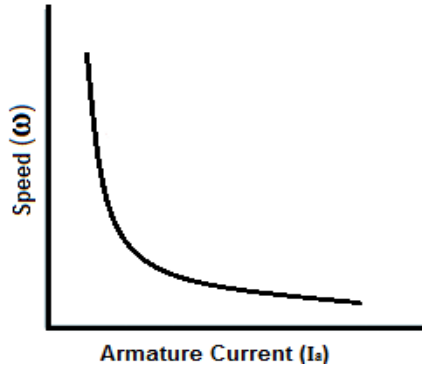
$$\omega = \frac{V}{K_1 I_a} - \frac{R_a + R_{se}}{K_1}$$

Torque-Current characteristics (T_a vs I_a)



$$T_a = K_1 I_a^2$$

Speed-Torque characteristics (ω vs T_a)



$$\omega = \frac{V}{K_1 I_a} - \frac{R_a + R_{se}}{K_1}$$

APPLICATIONS

Shunt Motor

- Wiper
- Automatic windscreen
- Drills
- Conveyors
- Fans
- Boring mills
- Shapers
- Blowers
- Spinning and weighing machine
- Centrifugal pumps

Series Motor

- Cranes
- Air compressor
- Lifts
- Elevators
- Winching system
- Electric traction
- Hair drier
- Vacuum cleaner and in speed regulation application
- power tools
- Sewing machine
- Electric footing

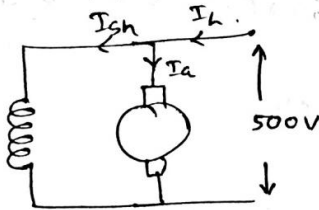
Compound Motor

- Presses
- Electric shovels
- Reciprocating machine
- Conveyors
- Stamping machine
- Elevators

- Compressors
- Rolling mills
- Heavy planners

16.

a)



At no-load

$$I_L = 5A$$

$$I_{sh} = \frac{500}{250} = 2A$$

$$I_a = I_L - I_{sh} = 3A$$

$$E_{b0} = V - I_a R_a = 500 - (3 \times 1.1) = \underline{496.7V}$$

$$N_0 = 1200 \text{ rpm}$$

At full-load

$$I_L = 30A$$

$$I_{sh} = 2A$$

$$I_a = I_L - I_{sh} = 28A$$

$$E_{b1} = V - I_a R_a = \cancel{497.2V} \quad 469.2V$$

$$N_1 = ?$$

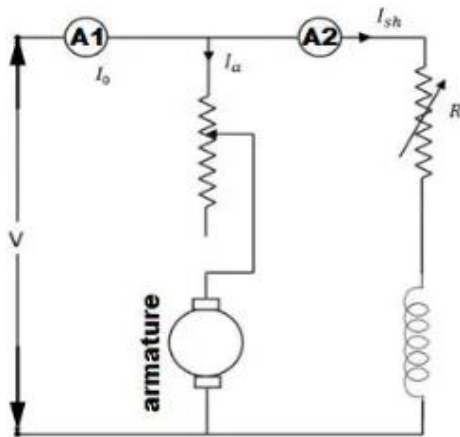
$$E_b \propto N$$

$$\begin{aligned} \frac{E_{b0}}{E_{b1}} &= \frac{N_0}{N_1} \implies N_1 = \frac{E_{b1}}{E_{b0}} \times N_0 \\ &= \frac{469.2}{496.7} \times 1200 \\ &= \underline{\underline{1133.56 \text{ rpm}}} \end{aligned}$$

b)

Swinburne's test is the most commonly used and simplest method of testing of shunt and compound wound DC machines which have constant flux. In this test the efficiency of the machine at any load is pre-determined. We can run the machine as a motor or as a generator. In this method of testing no load losses are measured separately and eventually we can determine the efficiency.

The circuit connection for Swinburne's test is shown in figure below. The speed of the machine is adjusted to the rated speed with the help of the shunt regulator R as shown in figure.



Calculation of Efficiency

Let, I_o is the no load current (it can be measured by ammeter A1)

I_{sh} is the shunt field current (it can be measured by ammeter A2)

Then, no load armature current = $(I_o - I_{sh})$

Also let, V is the supply voltage. Therefore, No load power input = VI_o watts.

In Swinburne's test no load power input is only required to supply the losses. The losses occur in the machine mainly are:

- Iron losses in the core
- Friction and windings losses
- Armature copper loss.

Since the no load mechanical output of the machine is zero in Swinburne's test, the no load input power is only used to supply the losses.

The value of armature copper loss = $(I_o - I_{sh})^2 R_a$

Here, R_a is the armature resistance.

Now, to get the constant losses we have to subtract the armature copper loss from the no load power input.

Then,

$$\text{Constant Losses, } W_c = VI_o - (I_o - I_{sh})^2 R_a$$

After calculating the no load constant losses now we can determine the efficiency at any load.

Let, I is the load current at which we have to calculate the efficiency of the machine.

Then, armature current (I_a) will be $(I - I_{sh})$, when the machine is motoring and $I_a = I + I_{sh}$ when the machine is generating.

Calculation of Efficiency when the machine works as Motor

Power Input = VI

Armature Copper Loss =

$$P_{CU} = I^2 R_a = (I - I_{sh})^2 R_a$$

Constant Loss =

$$W_C = VI_0 - (I_0 - I_{sh})^2 R_a$$

$$\text{Total losses} = P_{CU} + W_C$$

Thus,

Efficiency of motor,

$$\eta_m = \frac{\text{output}}{\text{input}} = \frac{\text{input} - \text{losses}}{\text{input}} = \frac{VI - (P_{CU} + W_C)}{VI}$$

Calculation of Efficiency when the machine works as Generator

Power Input = VI

Armature Copper Loss =

$$P_{CU} = I_a^2 R_a = (I + I_{sh})^2 R_a$$

Constant Loss =

$$W_C = VI_0 - (I_0 - I_{sh})^2 R_a$$

$$\text{Total losses} = P_{CU} + W_C$$

Thus,

Efficiency of generator,

$$\eta_g = \frac{\text{output}}{\text{input}} = \frac{\text{input} - \text{losses}}{\text{input}} = \frac{VI - (P_{CU} + W_C)}{VI}$$

Advantages of Swinburne's Test:

- This test is very convenient and economical as it is required very less power from supply to perform the test.
- Since constant losses are known, efficiency of Swinburne's test can be pre-determined at any load.

Disadvantages of Swinburne's Test

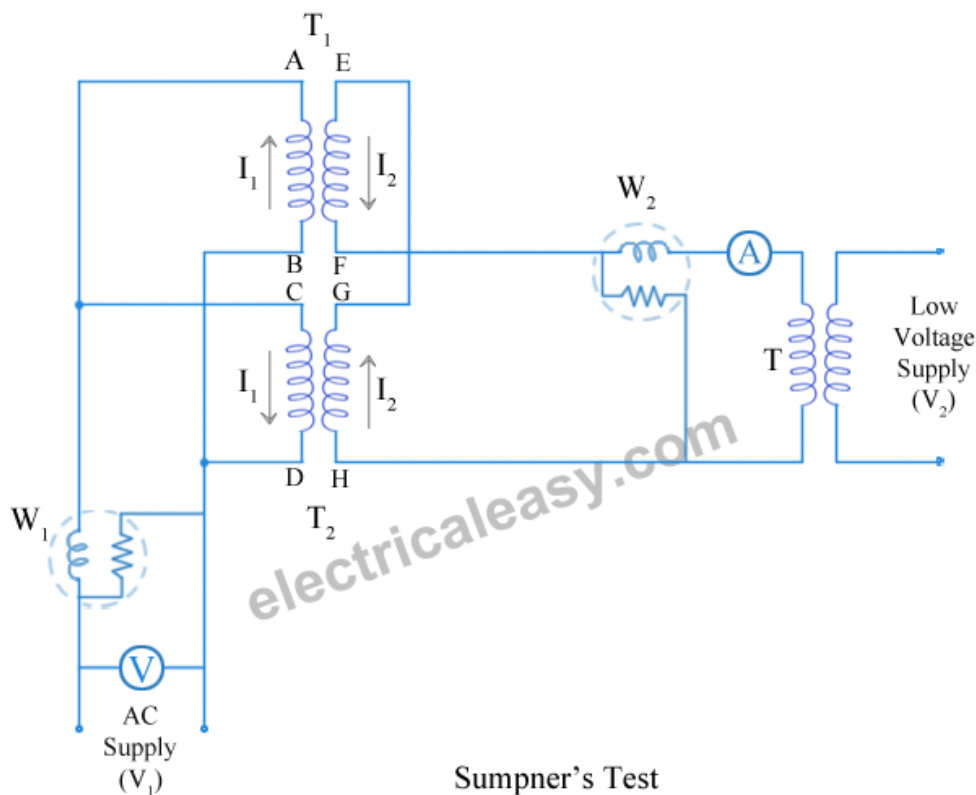
- Iron loss is neglected though there is change in iron loss from no load to full load due to armature reaction.

- We cannot be sure about the satisfactory commutation on loaded condition because the test is done on no-load.
- We can't measure the temperature rise when the machine is loaded. Power losses can vary with the temperature.
- In DC series motors, the Swinburne's test cannot be done to find its efficiency as it is a no load test.

17.

a) **Sumpner's Test**

Sumpner's test or back to back test can be employed only when two identical transformers are available. Both transformers are connected to supply such that one transformer is loaded on another. Primaries of the two identical transformers are connected in parallel across a supply. Secondaries are connected in series such that emf's of them are opposite to each other. Another low voltage supply is connected in series with secondaries to get the readings, as shown in the circuit diagram shown below.



In above diagram, T_1 and T_2 are identical transformers. Secondaries of them are connected in voltage opposition, i.e. EEF and EGH. Both the emf's cancel each other, as transformers are identical. In this case, as per superposition theorem, no current flows through secondary. And thus the no load test is simulated. The current drawn from V_1

is $2I_0$, where I_0 is equal to no load current of each transformer. Thus input power measured by wattmeter W_1 is equal to iron losses of both transformers.

i.e. iron loss per transformer $P_i = W_1/2$.

Now, a small voltage V_2 is injected into secondary with the help of a low voltage transformer. The voltage V_2 is adjusted so that, the rated current I_2 flows through the secondary. In this case, both primaries and secondaries carry rated current. Thus short circuit test is simulated and wattmeter W_2 shows total full load copper losses of both transformers.

i.e. copper loss per transformer $P_{Cu} = W_2/2$.

From above test results, the full load efficiency of each transformer can be given as:

$$\% \text{ full load efficiency of each transformer} = \frac{\text{output}}{\text{output} + \frac{W_1}{2} + \frac{W_2}{2}} \times 100$$

b)

$$\eta = \frac{x \cdot VA \cos \phi}{x \cdot VA \cos \phi + W_i + W_{Cu} x^2}$$

$$(i) \quad x = 1, \cos \phi = 1$$

$$W_i = \underline{70 \text{ W}} \quad (\text{oc test})$$

$$W_{Cu} \text{ at } 10 \text{ A} = \underline{60 \text{ W}}$$

from lv side,

$$\text{full load current, } I_1 = \frac{4 \times 10^3}{200} = 20 \text{ A}$$

$$\therefore W_{Cu} \text{ at } 20 \text{ A} = 60 \times \left(\frac{20}{10}\right)^2 = \underline{\underline{240 \text{ W}}}$$

$$\therefore \eta = \frac{4000 \times 1}{4000 + 70 + 240} = \underline{\underline{91.16\%}}$$

$$\text{Secondary voltage} = \left[\frac{E_2}{I_2} r_{02} \cos \phi + I_2 x_{02} \sin \phi \right]$$

To find r_{02} & x_{02} from SC test

$$Z_{02} = \frac{V_{sc}}{I_{sc}} = \frac{20}{10} = 10 \Omega$$

$$I^2 r_{02} = W_{sc}$$

$$r_{02} = \frac{60}{10^2} = 0.6 \Omega$$

$$x_{02} = 9.98 \Omega$$

$$\therefore \text{Secondary voltage, } V_2 = 400 - 6 = \underline{394 \text{ V}}$$

iv) Full load at 0.8 pf lag.

$$\eta = \frac{x \text{ VA } \cos \phi}{x \text{ VA } \cos \phi + W_i + x^2 W_{cu-FL}}$$

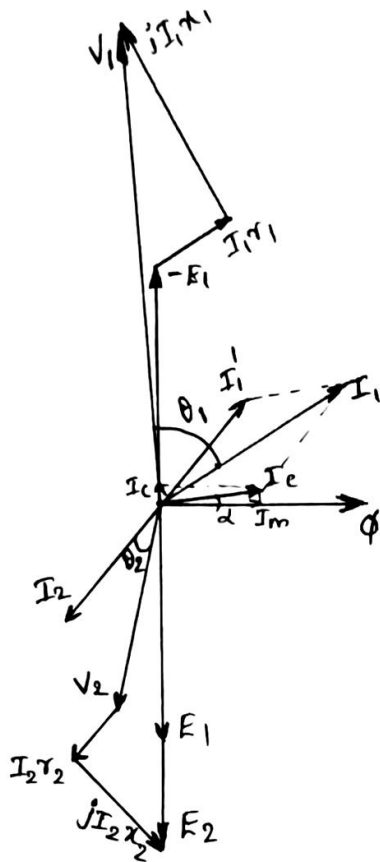
$$= \frac{4000 \times 0.8}{4000 \times 0.8 + 70 + 240} = \underline{91.17 \%}$$

$$V_2 = 400 - [10 \times 0.6 \times 0.8 + 10 \times 9.98 \times 0.6]$$

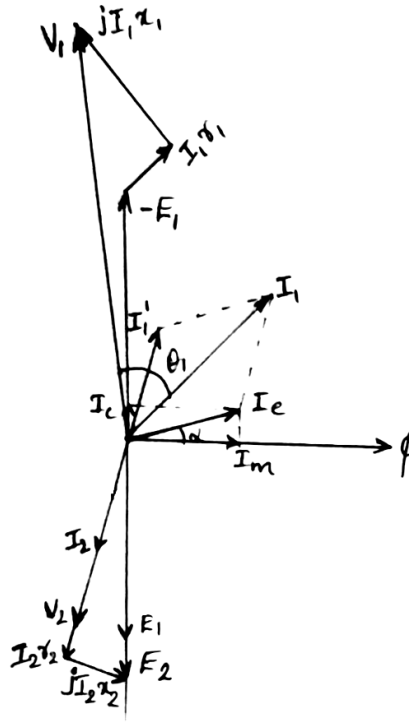
$$= \underline{335.32 \text{ V}}$$

18.

a) (i) Inductive Load (LPF)



(ii) Resistive Load (UPF)



b)

At maximum efficiency

$$\text{copper losses} = \text{core losses} = W$$

$$\eta_{\max} = \frac{\text{output power}}{\text{output power} + \text{losses}}$$

$$0.98 = \frac{15000}{15000 + 2W} \Rightarrow W = 153.06 \text{ W}$$

$$\therefore \text{Core losses} = \text{copper losses} = \underline{\underline{153.06 \text{ W}}}$$

$$\begin{aligned}\% \text{ All-day efficiency} &= \frac{\text{output in kWhr}}{\text{Input in kWhr}} \times 100 \\ &= \frac{\text{output in kWhr}}{\text{output in kWhr} + \text{losses in kWhr}} \times 100\end{aligned}$$

$$\begin{aligned}\text{output in kWhr} &= 2 \times 12 + 12 \times 6 + 18 \times 6 \\ &= \underline{\underline{204 \text{ kWhr}}}\end{aligned}$$

$$\begin{aligned}\text{Core losses in kWhr} &= 153.06 \times 10^{-3} \times 24 \\ &= \underline{\underline{3.673 \text{ kWhr}}}\end{aligned}$$

Copper losses in kWhr

load (kW)	load (kVA) = load (kW)/pf	Copper loss at that load (kW)	Copper loss in kWhr (kW × time)
2 kW, 0.5 pf	4 kVA	$0.15306 \times \left(\frac{4}{15}\right)^2 = 0.0109 \text{ kW}$	0.1308
12 kW, 0.8 pf	15 kVA	0.15306 kW	0.91836
18 kW, 0.9 pf	20 kVA	0.2721 kW	1.6326

$$\text{total copper losses in kWhr} = 2.68176 \text{ kWhr}$$

$$\begin{aligned}\therefore \% \eta_{\text{all-day}} &= \frac{204}{204 + 2.68176 + 3.673} \times 100 \\ &= \underline{\underline{96.98\%}}\end{aligned}$$

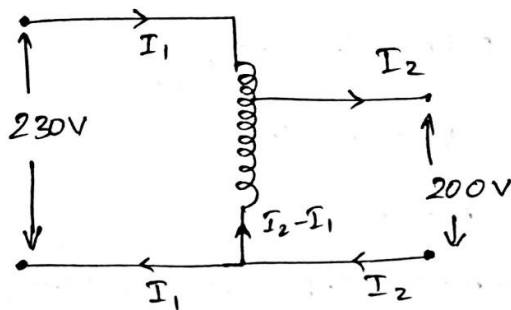
19.

a)

Conditions for Paralleling

- The line voltage ratio of the transformers must be same.
- The transformers should have equal per unit impedance.
- The transformers should have the same polarity.
- There should not be any phase difference between the secondary line voltages of all the transformers.
- The phase sequence of secondary line voltages of all the transformers should be same.

b)



load = 20 kW at upf.

$$\therefore I_2 = \frac{20 \times 10^3}{200} = \underline{\underline{100 \text{ A}}}$$

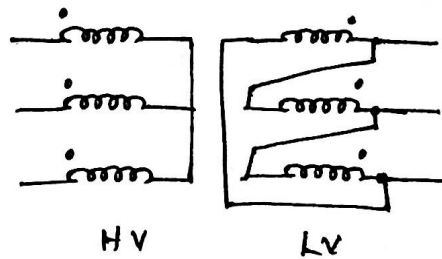
$$I_1 = \frac{20000}{230} = \underline{\underline{86.96 \text{ A}}}$$

$$I_2 - I_1 = \underline{\underline{13.04 \text{ A}}}$$

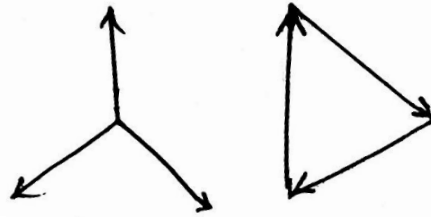
20.

a) **Yd11**

- HV side is star connected whereas LV side is delta connected.
- Voltage at LV side leads the voltage at HV side by 30°.



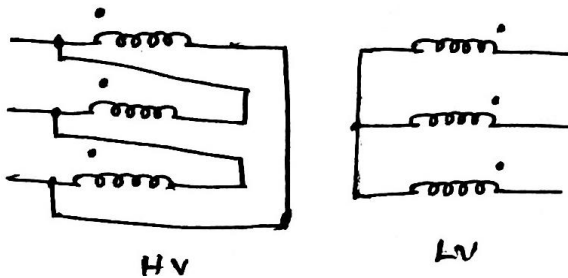
Circuit Diagram



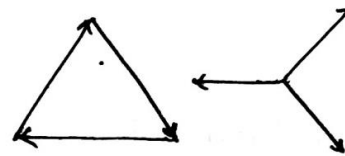
Phasor Diagram

Dy1

- HV side is delta connected whereas LV side is star connected.
- Voltage at LV side lags the voltage at HV side by 30° .



Circuit Diagram



Phasor Diagram

b)

- In some high rating transformers, a third winding is constructed in addition to the primary and secondary windings. This third winding is called Tertiary Winding and the transformer is called Three Winding Transformer.
- The voltage ratings of all three windings of transformer are usually unequal. The primary winding has the highest voltage rating; the tertiary has the lowest voltage rating, and the secondary has the intermediate voltage rating.
- The most significant advantage of the third winding is that the harmonic generated by the primary and secondary winding extinguish by the third winding. The third winding is connected in delta.

